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Inventor(s)	Rains; Gerald Eric et al.

Agricultural equipment with acoustic/vibration sensor and signal processing system

Abstract

An acoustic or vibration sensor is disposed relative to a portion of an agricultural machine. A sensor signal generated by the acoustic or vibration sensor is provided to a signal processor that identifies acoustic signatures or vibration patterns to determine a machine operation characteristic responsive to the agricultural machine performing an operation. The machine operation characteristic can be a characteristic of the agricultural machine, the soil, and/or furrow opened by the agricultural machine.

Inventors: Rains; Gerald Eric (Eldridge, IA), McCartney; Scott C. (Bettendorf, IA), Hubner; Cary S. (Geneseo, IL), Lichtenwalner, Jr.; Houstin Lee (Emmaus, PA), Ferren; Bran (Los Angeles, CA)

Applicant: Deere & Company (Moline, IL)

Family ID: 89843667

Assignee: DEERE & COMPANY (Moline, IL)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9702753	12/2016	Johnson et al.	N/A	N/A
9779330	12/2016	Wellington et al.	N/A	N/A
9826683	12/2016	Burns et al.	N/A	N/A
9832928	12/2016	Dybro et al.	N/A	N/A
9964484	12/2017	Haiges et al.	N/A	N/A
10989833	12/2020	Ferren et al.	N/A	N/A
2012/0227647	12/2011	Gelinske et al.	N/A	N/A
2019/0141880	12/2018	Zemenchik	172/1	A01C 5/064
2019/0174667	12/2018	Gresch	N/A	A01B 61/044
2020/0132654	12/2019	Pomedli	N/A	N/A
2020/0236843	12/2019	Graham	N/A	A01C 5/064
2020/0404831	12/2019	Kowalchuk et al.	N/A	N/A
2022/0304212	12/2021	Glovier	N/A	A01B 61/042
2022/0391644	12/2021	Honkanen	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
114264457	12/2021	CN	G01N 3/56
2966966	12/2015	EP	N/A
WO 2014/197973	12/2013	WO	N/A
WO 2019/157521	12/2018	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report and Written Opinion issued in European Patent Application No. 24155768.5 dated Jul. 8, 2024, in 13 pages. cited by applicant

Extended European Search Report and Written Opinion issued in European Patent Application No. 24155772.7 dated Jul. 8, 2024, in 09 pages. cited by applicant

Primary Examiner: Gray; Francis C

Attorney, Agent or Firm: KELLY, HOLT & CHRISTENSON, P.L.L.C

Background/Summary

FIELD OF THE DESCRIPTION

(1) The present description relates to agricultural equipment. More specifically, the present description relates to a system that detects and processes acoustic or vibration signals on an agricultural machine.

BACKGROUND

(2) There are a wide variety of different types of agricultural equipment that can be used to plant seeds or apply other commodities to a field. Such equipment can include air seeders. Air seeders have an air cart with one or more central seed or commodity tanks. The seed or commodity in the tank is metered by a metering system into common product delivery tubes called “primaries”. The seed or commodity is delivered (using air delivery) to a tool for disbursement. The tool disburses the product into secondary tubes which deliver the product to individual furrows opened by the tool. The furrows are opened by a furrow opener and closed after seed is delivered to the furrow by a closer. Other agricultural equipment can include planters that have row units.

(3) The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

(4) An acoustic or vibration sensor is disposed relative to a portion of an agricultural machine. A sensor signal generated by the acoustic or vibration sensor is provided to a signal processor that identifies acoustic signatures or vibration patterns to determine a machine operation characteristic responsive to the agricultural machine performing an operation. The machine operation characteristic can be a characteristic of the agricultural machine, the soil, and/or furrow opened by the agricultural machine.

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a partial pictorial, partial block diagram illustrating an agricultural system in which an air seeder is towed by a tractor.

(2) FIG. 1B is a top view of one example of a planting machine, shown in partial pictorial, partial schematic form.

(3) FIG. 1C is a side view of one example of a row unit of the planting machine shown in FIG. 1B.

(4) FIG. 2 is a perspective view of an air cart.

(5) FIG. 3 is perspective view of metering devices and primary delivery tubes.

(6) FIG. 4A is a block diagram of one example of an agricultural system.

(7) FIG. 4B an illustration of one example of an acoustic/vibration sensor.

(8) FIGS. 5A and 5B (collectively referred to herein as FIG. 5) show a flow diagram illustrating one example of the operation of the agricultural system.

(9) FIG. 6 is a block diagram of one example of a remote server architecture.

(10) FIGS. 7-9 show examples of mobile devices that can be used in the systems and architectures shown in the previous FIGS.

(11) FIG. 10 is a block diagram of one example of a computing environment that can be used in the

systems and architectures shown in the previous FIGS.

DETAILED DESCRIPTION

(12) As discussed above, air seeders have an air cart with one or more tanks that hold material to be delivered to the field. The air cart also has a metering system corresponding to each tank. The metering system meters material from the corresponding tank into a primary delivery tube that, and a fan blows air through that tube in order to move the material through the primary delivery tube to a seeding tool. The seeding tool receives material from the primary delivery tube and disperses it into a plurality of secondary delivery tubes. Each secondary delivery tube delivers material to a particular furrow that is opened by a furrow opener on the tool. After the material is placed in the furrow, the furrow can be closed by a furrow closer on the tool.

(13) Blockages can occur both upstream of the metering system, (e.g., in the tank), and downstream of the metering system, (e.g., in the primary or secondary delivery tubes). Therefore, some current systems include a blockage sensor on the secondary tubes on the air tool in order to determine whether any material is passing through that secondary delivery tube. If material is not detected by the blockage sensor, then the blockage sensor generates a signal indicative of a blockage somewhere upstream of the blockage sensor.

(14) However, when the air cart has multiple tanks, those tanks can be configured to meter multiple different materials through a single primary delivery tube. Those multiple materials are simultaneously delivered by the primary delivery tube to the secondary delivery tubes. Therefore, the blockage sensors in the secondary tubes cannot indicate when a blockage occurs with respect to one material metered from one of the tanks, but not the other material. Instead, with current sensors, as long as any material is sensed by the blockage sensor, the sensor signal will not indicate a blockage. Similarly, the current blockage sensors cannot indicate whether the materials are being delivered through the primary and secondary delivery tubes in the correct ratio relative to one another. Current sensors only detect whether material is flowing through the secondary tubes.

(15) In addition, air seeders can encounter other problematic situations as well. For instance, after a furrow is opened, there may still be a large amount of residue or other debris in the furrow. Also, if the soil is muddy, the furrow opener or the furrow closer (which are often rotating discs) can become stuck so that they are no longer rotating, but are instead simply being dragged through the mud.

(16) It can thus be seen that there are a wide variety of different types of blockage characteristics, furrow/soil characteristics, or machine characteristics, that can present problems.

(17) The present description thus proceeds with respect to a system that includes one or more acoustic or vibration sensors that sense sounds or vibrations and generate a sensor signal indicative of those sounds or vibrations. An advanced data processing system, such as Fast Fourier Transforms (FFTs) and windowing or level crossing thresholds, an artificial intelligence signal processor (e.g., a convolutional neural network) or other machine learning processor is used to generate an output, based upon the acoustic or vibration sensor signals indicative of one of a plurality of different characteristics. For instance, in one example the artificial neural network can identify acoustic signatures of the different types of material traveling through the secondary delivery tube to identify whether, and how much of, each different type of material is traveling through that tube. This can be used to determine whether a blockage exists, or whether a partial blockage exists and, if so, which meter or tank contains the blockage or partial blockage. Similarly, the acoustic or vibration sensors can be mounted to the tool to detect furrow or soil characteristics. By way of example, when the furrow opener or furrow closer engages different materials on the field surface or in the soil, those different materials will generate a different acoustic pattern or signature. Normal soil, for instance, may generate one acoustic signature, while soil that has a relatively large amount of residue (e.g., from a previous planting season) may generate a different acoustic signal as the tool engages that soil. Further, the acoustic or vibration sensors can be used to sense sounds or vibrations that may indicate machine characteristics. By way of example, if a

furrow closer is a disc closer that normally rotates to push dirt back over the furrow, then the sound that the bearings make (as the disc is rotating) will generate one type of acoustic pattern or signature. However, if the soil is muddy so that the disc closer is no longer rotating, then the acoustic pattern or signature corresponding to the bearings will no longer be present. Thus, the artificial intelligence signal processor can process the acoustic or vibration signals to generate outputs indicative of blockage characteristics, furrow/soil characteristic, and/or machine characteristics.

(18) The present description will proceed with respect to a material application tool being an air seeder that has an air cart and a seeding tool. The air cart has a plurality of tanks, each with a corresponding meter. The air cart also has a delivery system that delivers metered material to different work points on a seeding tool where furrows are opened by openers on the seeding tool and closed by closers. The present description also proceeds with respect to a planter and/or row unit. It will be noted that the material application tool can be an implement that seeds and applies fertilizer or that applies other material as well.

(19) FIG. 1A is a side view of an example of an agricultural system **100** which includes an agricultural implement, in particular an air or pneumatic seeder **102**. In the example shown in FIG. 1A, the seeder **102** comprises a tilling implement (or seeding tool) **104** (also sometimes called a drill) towed between a tractor (or other towing vehicle) **106** and a commodity cart (also sometimes called an air cart) **108**. The commodity cart **108** has a frame **110** upon which a series of product tanks **112**, **114**, **116**, and **118**, and wheels **120** are mounted. Each product tank has a door (a representative door **122** is labeled) releasably sealing an opening at the upper end of the tank for filling the tank with product, most usually seed or a commodity of one type or another. A metering system **124** is provided at a lower end of each tank (a representative one of which is labeled in FIG. 1A and others are shown below) for controlled feeding or draining of product (most typically granular material) into a pneumatic distribution system **126**. The tanks **112**, **114**, **116**, and **118** can hold, for example, a material or commodity such as seed or fertilizer to be distributed to the soil. The tanks can be hoppers, bins, boxes, containers, etc. The term “tank” shall be broadly construed herein. Furthermore, one tank with multiple compartments can also be provided instead of separate tanks.

(20) The tilling implement **104** includes a frame **128** supported by ground engaging wheels **130**. Frame **128** is connected to a leading portion of the commodity cart **108**, for example by a tongue style attachment (not labeled). The commodity cart **108** as shown is sometimes called a “tow behind cart,” meaning that the cart **108** follows the tilling implement **104**. In an alternative arrangement, the cart **108** can be configured as a “tow between cart,” meaning the cart **108** is between the tractor **106** and tilling implement **104**. In yet a further possible arrangement, the commodity cart **108** and tilling implement **104** can be combined to form a unified rather than separated configuration. These are just examples of additional possible configurations. Other configurations are even possible and all configurations should be considered contemplated and within the scope of the present description.

(21) In the example shown in FIG. 1A, tractor **106** is coupled by couplings **103** to seeding tool **104** which is coupled by couplings **105** to commodity cart **108**. The couplings **103** and **105** can be mechanical, hydraulic, pneumatic, and electrical couplings and/or other couplings. The couplings **103** and **105** can include wired and wireless couplings as well.

(22) The pneumatic distribution system **126** includes one or more fans located generally at **121** connected to a product delivery conduit structure having multiple product flow passages **132** referred to as primary delivery tubes **132**. The fan directs air through the primary delivery tubes **132**. Each product metering system **124** controls delivery of product from its associated tank at a controllable rate to the transporting airstreams moving through primary delivery tubes **132**. In this manner, each flow passage defined by each primary delivery tube **132** carries product from one or more of the tanks to a secondary distribution tower **134** on the tilling implement **104**. Typically,

there will be one tower **134** for each primary delivery tube **132**. Each tower **134** includes a secondary distributing manifold **136**, typically located at the top of a vertical tube. The secondary distributing manifold **136** divides the flow of product into a number of secondary distribution tubes **138**. Each secondary distribution tube **138** delivers product to one of a plurality of ground engaging tools **140** (also known as ground openers) that define the locations of work points on tilling implement **104**. The ground engaging tools (or furrow openers) **140** open a furrow in the soil **144** and facilitate deposit of the product therein. The number of primary delivery tubes **132** that feed into secondary distribution may vary from one to eight or ten or more, depending at least upon the configuration of the commodity cart **108** and tilling implement **104**. Depending upon the cart and implement, there may be two distribution manifolds **136** in the air stream between the meters **124** and the ground engaging tools **140**. Alternatively, in some configurations, the product is metered directly from the tank or tanks into secondary distribution lines that lead to the ground engaging tools **140** without any need for an intermediate distribution manifold. The product metering system **124** can be configured to vary the rate of delivery of seed or other material to each work point on tilling implement **104** or to different sets or zones of work points on tilling implement **104**. The configurations described herein are only examples. Other configurations are possible and should be considered contemplated and within the scope of the present description.

(23) A firming or closing wheel **142** associated with each ground engaging tool **140** trails the tool and firms the soil over the product deposited in the soil. In practice, a variety of types of tools **140** are used including, but not necessarily limited to, tines, shanks and disks. The tools **140** are typically moveable between a lowered position engaging the ground and a raised position riding above the ground. Each individual tool **140** may be configured to be raised by a separate actuator. Alternatively, multiple tools **140** may be mounted to a common component for movement together. In yet another alternative, the tools **140** may be fixed to the frame **128**, the frame being configured to be raised and lowered with the tools **140**.

(24) Examples of air or pneumatic seeder **102** described above should not be considered limiting. The features described in the present description can be applied to any seeder configuration, or other material application machine, whether specifically described herein or not.

(25) FIG. 1A also shows that agricultural system **100** includes one or more acoustic or vibration sensors (referred to herein as acoustic/vibration sensors). In the example in FIG. 1A, the acoustic/vibration sensors are labelled **146**, **148**, **150**, **152**, and **154**. Acoustic/vibration sensors **146**, **148**, **150**, **152**, and **154** (also referred to herein as acoustic/vibration sensors **146-154** or sensors **146-154**) are configured to sense an acoustic parameter responsive to a sound and/or a vibration parameter responsive to a vibration. Acoustic/vibration sensors **146** and **148** are disposed within different secondary distribution tubes **138**. Acoustic/vibration sensors **150** and **152** are on the mounting structure for tools **140**, and acoustic/vibration sensor **154** is on the mounting structure for closing wheel **142**. It will be appreciated that there can be more or fewer acoustic/vibration sensors and those shown in FIG. 1A are shown for the sake of example only. FIG. 1A shows that agricultural system **100** can also include control system **156**. Control system **156** receives and processes signals from the acoustic/vibration sensors **146**, **148**, **150**, **152**, and **154** and generates an output based on the processed signals. Control system **156** is described in greater detail elsewhere herein.

(26) Also, it will be appreciated, that different portions of system **156** can reside on tractor **106**, on tool or implement **104**, and/or on air cart **108**, or all of the elements of system **156** can be located at one place (e.g., on tractor **106**). Elements of system **156** can be distributed to a remote server architecture or located or distributed in other ways as well.

(27) By way of example, acoustic/vibration sensors **146** and **148** can generate acoustic signals indicative of the acoustic pattern or signature or indicative of the vibration pattern or signature of material traveling through the secondary distribution lines **138**, to which sensors **146** and **148** are mounted. In one example, the acoustic signature or vibration signature of the material will be

different depending on the quantity of material and the type of material traveling through the secondary distribution line. Therefore, control system **156** can use machine learning (e.g., artificial intelligence, an artificial neural network, a convolutional neural network, etc.) to process the sensor signals generated by sensors **146** and **148** and identify the quantity and types of material traveling through the secondary distribution lines **138** based on the acoustic pattern or signature or vibration pattern or signature. As used herein, acoustic signature and acoustic pattern will be used interchangeably as will vibration signature and vibration pattern. Also, acoustic/vibration signature or pattern includes, without limitation, acoustic signature, acoustic pattern, vibration signature, and/or vibration pattern.

(28) Assume, for instance, that one of the secondary distribution lines **138** receives material from two different tanks **114** and **116**, that are both metered into a single primary distribution tube **132**. In that case, there may be a blockage in either or both of the tanks **116** and **118**, or the blockage could be at the metering devices **124** corresponding to the different tanks **116** and **118**, or below those metering devices. Therefore, there may be a full or partial blockage of material flow from one or both of the tanks **116** and **118**. By processing the acoustic signatures generated by sensors **146** and **148**, control system **156** can determine what types of material are flowing through the corresponding secondary distribution tubes **138**, and the quantity of each material and/or the ratio of the multiple different materials traveling through the same secondary distribution line **138** relative to one another. In this way, if one tank has a blockage but the other tank does not, then even if a secondary distribution line **138** is carrying a single material, the control system **156** can identify that flow of the other material is blocked, which tank is blocked, whether it is a partial or complete blockage, and/or which meter is blocked. The control system **156** can process the acoustic signal or vibration signal to identify any characteristics of the material to be delivered (e.g., to identify materials being delivered, whether a material is blocked, quantities or ratios of materials being delivered, or other parameter or characteristic that is responsive to delivery of material by the delivery system, etc.) that can be identified by the acoustic signature or pattern and/or the vibration signature or pattern. This information can be used to generate an output to the operator, or for a variety of other uses, some of which are described in greater detail elsewhere herein. Also, while the present discussion proceeds with respect to, in one example, having two different tanks that deliver two different materials to a single primary distribution tube **132**, it will be appreciated that any number N of tanks can be deliver N different materials to a single primary delivery tube **132**. Other configurations are also contemplated herein.

(29) Sensors **150** and **152** may be acoustic or vibration sensors mounted to sense the acoustic or vibration pattern generated by the corresponding tool **140** to which they are mounted. The acoustic pattern or the vibration pattern may differ depending upon the characteristics of the soil or furrow being engaged by the tools **140**. For instance, assume that a tool **140** is encountering soil that is relatively clear of residue from a prior growing season. The acoustic signature or vibration pattern generated by the corresponding sensor **150** or **152** may be a first signature or pattern. However, if the tool **140** is traveling through soil that has a relatively heavy concentration of residue from a prior growing season, then the acoustic signature or vibration pattern sensed by the corresponding sensor **150** or **152** will be a different signature or pattern. Further, if the tool **140** engages a rock or other obstacle, that will also be indicated by the acoustic signature or vibration pattern generated by the corresponding sensor. Also, where the soil is of one type (e.g., sandy) then the acoustic signature or vibration pattern may differ from the acoustic signature or vibration pattern where the soil is clay. Therefore, control system **156** can process the acoustic signal or vibration signal to identify any characteristics of the furrow or soil (e.g., clean soil, soil with residue, soil with rocks or other objects, soil type, soil moisture, etc.) that can be identified by the acoustic signature or vibration pattern.

(30) Sensor **154** may also be an acoustic sensor which generates a signal indicative of the acoustic pattern or signature or the vibration pattern or signature generated by a part of the machine, such as

closing wheel **142**. Control system **156** can generate an output indicative of machine characteristics corresponding to the part of the machine, such as closing wheel **142** based upon the acoustic signature or pattern. For instance, if wheel **142** is rotating, then the bearings in wheel **142** will influence the acoustic signature or vibration pattern sensed by sensor **154**. However, if the soil is muddy so that wheel **142** is no longer rotating but is instead being dragged through the muddy soil, then the vibration pattern or acoustic signature will be different, because there will be no bearing noise or vibration. Also, bearings or other moving parts may generate different acoustic signatures or vibration patterns as they wear. Further, if the tool is supposed to be in engagement with the ground, but the acoustic signature or vibration pattern is consistent with the tool being out of engagement with the ground, this can be indicative of a tool malfunction. Any type of machine characteristic that produces an acoustic signature or vibration pattern can be sensed. Therefore, control system **156** can identify machine characteristics based upon the acoustic signature or vibration pattern generated by sensor **154**. It will be appreciated that there are a wide variety of different types of blockage characteristics or other material flow characteristics, furrow/soil characteristics, and/or machine characteristics that can be identified based upon the outputs from the acoustic/vibration sensors, and those described herein are described for the sake of example only.

(31) The present description also proceeds with respect to an example in which the sensors **150**, **152** can be deployed on a row unit of a planter. FIG. **1B** is a partial pictorial, partial schematic top view of another example of an agricultural system **100** that includes agricultural planting machine **400**, towing vehicle **94** that is operated by operator **92**, control system **156**, and material application control system **413**. Systems **156**, **413** can be on one or more individual parts of machine **400**, centrally located on machine **400**, or on towing vehicle **94**. Operator **92** can illustratively interact with operator interface mechanisms **96** to manipulate and control vehicle **94**, systems **156**, **413**, and some or all portions of machine **400**.

(32) Machine **400** is a row crop planting machine that illustratively includes a toolbar **402** that is part of a frame **404**. FIG. **1B** also shows that a plurality of planting row units **406** are mounted to the toolbar **402**. Machine **400** can be towed behind towing vehicle **94**, such as a tractor. FIG. **1B** shows that material can be stored in a tank **407** and pumped through a supply line **411** so the material can be dispensed in or near the rows being planted. In one example, a set of devices (e.g., actuators) **409** is provided to perform this operation. For instance, actuators **409** can be individual pumps that service individual row units **406** and that pump material from tank **407** through supply line **411** so the material can be dispensed on the field. In such an example, material application control system **413** controls the pumps **409**. In another example, actuators **409** are valves or nozzles and one or more pumps **415** pump the material from tank **407** to valves or nozzles **409** through supply line **411**. In such an example, material application control system **413** controls valves or nozzles **409** by generating valve or nozzle control signals, e.g., on a per-seed basis, as described below. The present discussion proceeds by describing the actuator **409** as a valve or actuator. It will be noted that actuator **409** can be a pump, a nozzle, a valve, or another actuator or combination of actuators. The control signal for each actuator **409** can, in one example, be a pulse width modulated control signal. The flow rate through the corresponding valve **409** can be based on the duty cycle of the control signal (which controls the amount of time the valve is open and closed). The flow rate can be based on multiple duty cycles of multiple valves or based on other criteria. Further, the material can be applied in varying rates on a per-seed or per-plant basis. For example, fertilizer may be applied at one rate when the fertilizer is being applied at a location spaced from a seed location and at a second, higher, rate when the fertilizer is being applied closer to the seed location. These are examples only. Sensors **146-154** can be mounted on a structure or structures on machine **400** and processed by control system **156**. Sensor **150** is shown mounted to a shank and communicating with control system **156**, but sensors **146-154** can be mounted in any other locations as well. Sensors **146-154** can sense sound and/or other vibrations generated by operation

of row unit **406**. Based on the sensed sounds and/or other vibrations, control system **156** can identify machine characteristics or other characteristics that can be used to generate control signals, as is described in more detail elsewhere herein.

(33) FIG. **1C** is a side view of one example of a row unit **406**, with valve **409** and systems **156** and **413** shown as well. Actuator **409** is shown in five possible locations labeled as **409**, **409A**, **409B**, **409C**, **409D**, and **409E**, but other locations are possible as well. Row unit **406** also illustratively includes a seed hopper **427**, one or more disc openers **414**, a set of gauge wheels **416**, and a set of closing wheels **418**. Seeds from hopper **427** are fed into a seed meter **424**, e.g., by gravity or from a centralized commodity distribution system (e.g., employing pneumatic commodity distribution to each row unit). The seed meter **424** controls the rate at which seeds are dropped into a seed tube or other seed delivery system, such as an assistive seed delivery seed system **466** which may be a brush belt or flighted belt from hopper **427**. The seeds can be sensed in the furrow by seed sensor **422**, or seed sensor **423** in delivery system **466** or the seeds can be sensed by a seed sensor located elsewhere.

(34) In the example shown in FIG. **1C**, liquid material is passed, e.g., pumped or otherwise forced, through supply line **411** to an inlet end of actuator **409**. Actuator **409** is controlled by control system **413** to allow the liquid to pass from the inlet end of actuator **409** to an outlet end of actuator **409**.

(35) As liquid passes through actuator **409**, the liquid travels through an application assembly **417** from a proximal end (which is attached to an outlet end of actuator **409**) to a distal tip (or application tip) **419**, where the liquid is discharged into a trench, or proximate a trench or furrow **462**, opened by disc opener **414** (as is described in more detail below). The distal tip **419** can be located in one or more different locations on row unit **406**, some of which are labeled **419** and **419A**.

(36) Some parts of row unit **406** will now be discussed in more detail. First, it will be noted that there are different types of seed meters **424**, and the one that is shown is shown for the sake of example only. In one example, each row unit **406** need not have its own seed meter. Instead, metering or other singulation or seed dividing techniques can be performed at a central location, for groups of row units **406**. The metering systems can include finger pick-up discs and/or vacuum meters (e.g., having rotatable discs, rotatable concave or bowl-shaped devices), among others. The seed delivery system can be a gravity drop system (such as seed tube) in which seeds are dropped through the seed tube and fall (via gravitational force) through the seed tube and out an outlet end into the seed trench **462**. Other types of seed delivery systems may be or may include assistive systems **466**, in that they do not simply rely on gravity to move the seed from the metering system into the ground. Instead, such an assistive system **466** actively assists the seeds in moving from the meter to a lower opening, where the seeds exit or are deposited into the ground or trench. These can be systems that physically capture the seed and move it from the meter to the outlet end of the seed delivery system or they can be pneumatic systems that pump air through the seed tube to assist movement of the seed. The air velocity can be controlled to control the speed at which the seed moves through the delivery system.

(37) The seed delivery system shown in FIG. **1C** is an assistive seed delivery system **466**. Assistive seed delivery system **466** may have a seed sensor **423** disposed thereon. Assistive seed delivery system **466** captures the seeds as they leave seed meter **424** and moves the seeds in the direction indicated by arrow **468** toward furrow **462**. System **466** has an outlet end **470** where the seeds exit assistive system **466**, into furrow **462**, where the seeds reach their final resting location. Also, it should be noted that there can be more than one sensor **146-154**, more than one seed sensor **422**, **423**, sensors of different types, sensors at different locations, etc. **7**

(38) FIG. **3** shows that row unit **406** also includes row cleaner **425** which clears residue or other material ahead of the opener **414**.

(39) A downforce actuator **426** is mounted on a coupling assembly **428** that couples row unit **406** to

toolbar **402**. Actuator **426** can be a hydraulic actuator, a pneumatic actuator, a spring-based mechanical actuator, or a wide variety of other actuators. In the example shown in FIG. **1C**, a rod **430** is coupled to a parallel linkage **432** and is used to exert an additional downforce (in the direction indicated by arrow **434**) on row unit **406**. The total downforce (which includes the force indicated by arrow **434** exerted by actuator **426**, plus the force due to gravity acting on row unit **406**, and indicated by arrow **436**) is offset by upwardly directed forces acting on closing wheels **418** (from ground **438**) and on row cleaner **425** (from ground **438**) and disc opener **414** (again from ground **438**). The differential force (which may also be referred to herein as the downforce margin) acts on the gauge wheels **416**. The gauge wheels **416** can be set to control the depth of the furrow **462** opened by openers **414**. The load on gauge wheel **416** can be sensed by a gauge wheel load sensor, which may be located anywhere on row unit **406** where it can sense that load. A set of gauge wheel control arms (or gauge wheel arm) movably mount gauge wheels **416** to shank or frame **452** and control an offset between gauge wheels **416** and the discs in double disc opener **414**, to control planting depth.

(40) In operation, row unit **406** travels generally in the direction indicated by arrow **460**. The double disc opener **414** opens a furrow **462** in the soil **438**, and the depth of the furrow **462** is set by planting depth actuator assembly **454**, which, itself, controls the offset between the lowest parts of gauge wheels **416** and disc opener **414**. Seeds are moved by assistive seed delivery system **466** into the furrow **462** and closing wheels **418** close the furrow **462**, e.g., push soil back into the furrow **462**.

(41) Seeds can be sensed by seed sensor **423** in assistive delivery system **466**. When the seeds are placed into the furrow **462**, the seeds can be sensed by seed sensor **422**. The seed sensors can be placed in a variety of different locations on row unit **406**, or on different components of row unit **406**.

(42) Material application control system **413** illustratively receives a signal from one of the seed sensors. Material application control system **413** can also receive a ground speed signal indicative of a speed of movement of row unit **406**, and then determines when to actuate actuator **409** so that material being applied through application assembly **417** (and out distal tip **419** of application assembly **417**) will be applied at a desired location relative to the seed in trench or furrow **462** and/or based on other planting characteristics identified by material application control system **413** or elsewhere.

(43) By way of example, it may be that some material is to be applied directly on the seed. In that case, system **413** times the actuation of actuator **409** so that the applied material will be applied at the seed location. In another example, it may be desirable to apply some material at the seed location and also apply material for a predetermined distance on either side of the seed location. In that case, system **413** generates the signal used to control actuator **409** so that the material is applied in the desired fashion. In other examples, it may be that the material is to be applied at a location between seeds in furrow **462**. By way of example, relatively high nitrogen fertilizer may be most desirably applied between seeds, instead of directly on the seed. In that case, system **413** is illustratively programmed with the desired location of the applied material relative to seed location, so that system **413** can determine when to actuate actuator **409** in order to apply the material between seeds. Further, as discussed above, actuator **409** can be actuated to dispense material at a varying rate. System **413** can control multiple actuators **409** to apply different material relative to different seeds, to aim a nozzle through which the material travels or to otherwise aim the material relative to the seed location, or to control the shape of the spray existing the tip **419**. As another example, it may be that two or more different types of seed are being planted and the seed sensor generates an output indicative of the location of the sensed seed and the type of seed detected. System **413** can then generate control signals to apply a first type of material (or quantity of material) to seeds of a first type and a second type of material (or quantity of material) to seeds of a second type. In another example, system **413** can control actuator **409** to dispense more material on

the seed location and less material at locations spaced from the seed location, or vice versa, or according to other patterns.

(44) FIG. 2 is a pictorial illustration of another example of an air cart **160**. Air cart **160** is similar to air cart **108** in FIG. 1, except that air cart **160** has three tanks **162**, **164**, and **166**. Each tank has a corresponding metering system **168**, **170**, and **172** that meters material from the corresponding tanks into one or more of the primary delivery tubes **132**.

(45) FIG. 3 shows a portion of air cart **160** with the tanks, wheels, and other portions removed, to show metering systems **168**, **170**, and **172** more clearly. FIG. 3 shows that material from tank **162** is metered by meter **168** into one or more of the primary delivery tubes **132**. Similarly, material from tank **164** is metered by metering system **170** into the primary delivery tubes **132**, and material from tank **166** is metered by metering system **172** into the primary delivery tubes **132**. Any number N of metering systems can meter any number of different materials into the same primary delivery tube **132**. In one example, a plurality of the metering systems **168**, **170**, and **172** can meter material into the same primary delivery tube **132**. For instance, in one example, material from tanks **162** and **164** can be metered by metering systems **168** and **170** into the same primary delivery tube **132**, while material from tank **166** is metered by metering system **172** into a separate primary delivery tube **132**. Therefore, the primary delivery tubes **132** may carry a single type of material, or a plurality of different types of material. It will also be noted that blockages may occur within the tanks **162**, **164**, or **166**, or within the metering systems **168**, **170**, or **172**, or on either side of the metering systems (e.g., on the input side from the tanks or on the output side to the primary delivery tubes).

(46) In accordance with the present example, the acoustic/vibration sensors in the secondary delivery lines **138** generate an acoustic or vibration signal that can be used by control system **156** to identify whether the proper materials are moving through the primary delivery tubes **132** and secondary delivery lines **138**, and whether the materials are in the proper proportion relative to one another. Therefore, if, for instance, acoustic/vibration sensor **146** is supposed to be sensing two different materials, but only generates an acoustic signal that has an acoustic signature indicative of a single material, then control system **156** can identify that the second material is missing from the secondary delivery line **138**. Thus, the primary delivery tube **132** that services that secondary delivery line **138** will also be missing that material (since it is not reaching line **138**). Thus, control system **156** can identify the particular meter or tank or line **138** where a blockage has occurred.

(47) FIG. 4A is a block diagram showing one example of a portion of agricultural system **100**. Some items shown in FIG. 4A are similar to those shown in previous figures, and they are similarly numbered. In the example shown in FIG. 4A agricultural system **100** includes one or more processors or servers **180**, data store **182**, communication system **184**, sensors **186**, system functionality **188** (which can include such things as metering systems, down pressure—or downforce—actuators, a propulsion system, a steering system, seed metering system(s), seed delivery systems(s), etc.), control system **156**, operator interface system **190**, and a wide variety of other items **192**. Sensors **186** can include a position sensor **194** as well as one or more acoustic/vibration sensors, such as sensors **146**, **148**, **150**, **152**, and **154** discussed above. Sensors **186** can include seed sensors **422**, **423**, and/or a wide variety of other sensors **196** as well. The sensors **186** generate sensor signals **198** and provide those sensor signals to control system **156**. Control system **156** can include signal processor training system **200**, signal conditioning system **202**, signal processing system (e.g., an advanced signal processing system such as FFTs with windowing or level crossing thresholds, a machine learning system, such as AI, convolutional neural network, etc.) **204** (signal processing system **204**), control signal generator **206**, and other items **208**. Signal processing system **204** can include blockage characteristic processor **210**, furrow/soil characteristic processor **212**, machine characteristic processor **214**, and other items **216**. Control signal generator **206** is shown generating a control signal **218**.

(48) FIG. 4A also shows that an operator **220** can interact with operator interface system **190** to

operate some or all of the portions of agricultural system **100**. For instance, operator **220** may be in an operator's compartment of tractor **106, 94**, or elsewhere.

(49) FIG. **4A** also shows that agricultural system **100** can be connected over network **222** to other systems **224** or other machines **226**. Therefore, network **222** may be a wide area network, a local area network, a near field communication network, a Wi-Fi network, a Bluetooth network, a cellular communication network, or any of a wide variety of other networks or combinations of networks. Other systems **224** can include farm manager systems, manufacturer systems, vendor systems, or a wide variety of other systems. Other machines **226** can include other machines operating in the same field as tractor **106, 94**, machines operating in different fields, tender vehicles, or machines performing subsequent operations in the present field or other machines.

(50) Before describing the operation of agricultural system **100** in more detail, a description of some of the items in system **100**, and their operation, will first be provided.

(51) Communication system **184** enables communication of the items in agricultural system **100** with one another. Therefore, system **184** may be a controller area network (CAN) bus and bus controller, or other communication systems. Further, communication system **184** may enable communication over network **222** to communicate with other systems **224**, other machines **226**, etc. Therefore, communication system **184** may vary depending on the type of network **222** over which it is to communicate.

(52) Operator interface system **190** can include mechanisms **96** (from FIG. **1B**) or any of a wide variety of operator interface mechanisms and corresponding logic used to control those mechanisms. For instance, operator interface system **190** can include a steering wheel, joysticks, levers, linkages, a microphone and speakers (e.g., where speech recognition and/or speech synthesis is provided), a display screen, a touch sensitive display screen, or other items. Where a display screen is used, user actuatable input mechanisms can be displayed to display user information and receive user inputs. Those mechanisms can include such things as icons, links, buttons, etc. The mechanisms can be actuated using a point and click device, touch gestures, etc. System **190** can include any of a wide variety of other audio, visual, and/or haptic output/input devices.

(53) Position sensor **194** can be a global navigation satellite system (GNSS) receiver, a cellular triangulation system, a dead reckoning system, or any of a wide variety of other sensors that sense the position of agricultural system **100** in a global or local coordinate system. Acoustic/vibration sensors **146, 148, 150, 152, and 154** (sensors **146-154**) can be similar to one another or different. For instance, the sensors **146-154** can be microphone sensors that sense audio sounds, such as grain striking a grain strike sensor, noises corresponding to bearings, stethoscope impact plate sensors (one of which is described below with respect to FIG. **4B**), piezoelectric sensors, inertial measurement units (IMUs), etc. By way of example, sensors **146-154** can include a piezoelectric sensor element that continuously captures a vibration amplitude with respect to time. The force acting on the piezoelectric element may be the force of grain or other material impacting the piezoelectric sensor element. The piezoelectric sensor element may convert the vibrations into corresponding voltage amplitude, continuously with respect to time, to generate a time varying amplitude wave form signal as an acoustic data stream. Other sensors can be used as well. The sensors **186** generate sensor signals **198** and provide those signals to control system **156**.

(54) Signal conditioning system **202** may perform signal conditioning on the sensor signals **198**. Such signal conditioning may include amplification, linearization, normalization, filtering, etc.

(55) Signal processing system **204** receives the conditioned signals and generates an output indicative of sensed parameters or characteristics. For instance, the input to signal processing system **204** may be an acoustic or vibration signal. Signal processing system **204** may, itself, be a convolutional neural network, or another type of machine learning or artificial intelligence or advanced processing system. Signal processing system **204** may identify features in the acoustic or vibration signal and classify those features to provide an output signal indicative of certain

characteristics represented by the features in the acoustic or vibration signal. By way of example, blockage characteristic processor **210** may be trained to identify the acoustic signatures of different types of material that may be sensed by sensors **146-154**. When those signatures are missing from the acoustic signal, blockage characteristic processor **210** may generate an output indicating that flow of a certain material is not reaching the corresponding sensors. Similarly, the signatures or patterns in the acoustic or vibration signal may vary based on the amount of each material sensed by the sensor. Therefore, the blockage characteristic processor **210** may be trained to recognize the amounts of each material, based on the acoustic signature or vibration patterns in the acoustic/vibration signals. Thus, the blockage characteristics generated by blockage characteristic processor **210** may also indicate the ratios of the different materials being sensed by the sensor. The blockage characteristics can be other characteristics that identify materials, blockages, partial blockages, which meters, tanks, or delivery tubes are blocked, material quantity and/or material ratios, etc.

(56) Furrow/soil characteristic processor **212** may receive an acoustic or vibration signal that captures acoustic or vibration features (e.g., acoustic signatures or vibration patterns) corresponding to the tillage implement **104** or machine **400** traveling through soil. Such features may include an acoustic or vibration signature that may have one form when the soil is relatively clear of obstructions, and may have another form when the soil contains residue, rocks, or other objects. The acoustic or vibration signatures may vary based on the soil type, soil moisture, or other soil or furrow characteristic. Thus, furrow/soil characteristic processor **212** can be trained to extract acoustic or vibration features (e.g., patterns and/or signatures) and generate an output, based upon those features, indicative of characteristics of the furrow and/or soil.

(57) Machine characteristic processor **214** may receive an acoustic signal or vibration signal and extract features from that signal indicative of characteristics of the machine (such as whether bearings are rotating, whether the seed delivery systems and seed metering systems or other machine parts are operating properly, whether the machine is in a proper position—e.g., in engagement with the soil, whether parts are worn, the degree of wear, etc.). Machine characteristic processor **214** can then generate an output indicative of the characteristics of the machine identified by the extracted features of the acoustic or vibration signal.

(58) Signal processor training system **200** can be used to train signal processing system **204** to identify the particular characteristics that are being identified. For instance, the signal processor training system **200** may be a supervised or unsupervised, machine learning system that trains signal processing system **204** based upon labeled training data or unlabeled training data. As an example, the training data may represent conditions that are to be sensed by blockage characteristic processor **210**, furrow/soil characteristic processor **212**, machine characteristic processor **214**, or other processing systems.

(59) Control signal generator **206** can then generate control signals based upon the blockage characteristics, the furrow/soil characteristics, and/or the machine characteristics output by signal processing system **204**. For instance, the control signal **218** can control operator interface system **190** to generate an alert or warning output to operator **220**. The alert may identify, for example, which tank or meter is blocked, what type of equipment is malfunctioning or is about to wear out, an indication that a furrow contains an undesirable amount of residue, etc. Control signal generator **206** can also generate a control signal to provide a geographic tag (geotag) to the characteristic so that the location where the characteristic was sensed can be stored and processed in other ways. By way of example, assume that furrow/soil characteristic processor **212** generates an output indicating that the tillage implement **104** has struck a rock. Control signal generator **206** can generate a control signal that outputs an indication of the rock strike, along with its geographic location, so that subsequent operations in the field can be controlled to avoid hitting the rock. Signal processing system **204** may also correlate sensed characteristics to the machine, material, and/or environmental conditions by storing metadata representing the machine, material and/or

environmental conditions along with the detected characteristics. Control signal generator **206** may generate a control signal to store the identified characteristics locally in data store **182** and/or remotely in another system **224**. Control signal generator **206** can generate a control signal to control communication system **184** to communicate the identified characteristics to other systems **224** or other machines **226** over network **222**. Control signal generator **206** can generate a stop signal (such as for an autonomous machine), a down force control signal to change down force on a row unit, or other control signals. The control signal can take a wide variety of other forms as well.

(60) Before describing the operation of agricultural system **100** in more detail, a description of one example of an acoustic blockage sensor will be described. FIG. **4B** shows one example of an acoustic blockage sensor **250**. Sensor **250** can be one or more of the acoustic/vibration sensors **146**, **148**, **150**, **152**, and/or **154**. In one example, sensor **250** includes a material input opening **252** and a material output opening **254**. Material to be sensed flows in through input opening **252** and contacts stethoscope impact plate **256**. In one example, in which sensor **250** is disposed in a secondary line **138** (shown in FIG. **1**), the material passing through the secondary lines **138** is directed through the input opening **252** of acoustic blockage sensor **250** and impacts the stethoscope impact plate **256**. The sound generated by that impact transitions through a conduit **258** (which may be a rubber tube or another type of tube). The material is then deflected to the output opening **254** of acoustic blockage sensor **250** where the material continues to flow through the secondary line **138**. The sound on stethoscope impact plate **256** is transmitted through conduit **258** to a microphone input on control system **156**.

(61) In one example, there may be multiple materials traveling through sensor **250**, and those materials will have different densities. Thus, each of the different materials create a different sound when impacting the stethoscope impact plate **256**. Therefore, in one example, signal processor training system **200** trains signal processing system **204** to recognize the acoustic signature generated by each material and different amounts of each material, as each material is sent through the acoustic sensor **250**. In one example, a plurality of different materials can be sent through sensor **250** at the same time, in different conditions, in different ratios relative to one another, etc. so training system **200** can train signal processing system **204** on those conditions and ratios as well. This allows the signal processing system **204** to learn the sound signature of each product, and of different product combinations. Once the learning has been completed, then signal processing system **204** can differentiate the flow of the different materials through acoustic blockage sensor **250**, even when multiple different materials are sent through the sensor **250** at the same time. Blockage characteristic processor **210** can, for example, identify the materials, which material is blocked or partially blocked and, by knowing which tank and meter is providing each material, identify the particular tank or meter where the blockage is occurring.

(62) FIG. **5** is a flow diagram illustrating one example of the operation of agricultural system **100**. It is first assumed that an agricultural planting machine is configured with acoustic/vibration sensors and with a trained signal processing system, as indicated by block **260** in the flow diagram of FIG. **5**. The agricultural planting machine can be an agricultural planting machine **400** with row units **406** as indicated by block **261**, an air seeder **262** or any of a wide variety of other planting machines. In one example, the planting machine can provide multiple materials through a primary tube, and through secondary tubes, through meters on an air cart, as indicated by block **264**. The planting machine can have a furrow opener **266**, and the acoustic/vibration sensors can be one or more microphones **268**. The acoustic/vibration sensors can be stethoscope or piezo sensors **270**, inertial measurement units **272**, or other sensors. The planting machine can include a signal processing system which may be an advanced signal processing system such as a trained machine learning, artificial intelligence processing system **274**, which may include such things as a convolutional neural network, etc. The acoustic/vibration sensors may be mounted to the secondary conduits, as indicated by block **276**. The acoustic/vibration sensors may be mounted to the furrow opener blades or the mounting structure for the furrow opener blades, or elsewhere on a row unit

406 as indicated by blocks **278**. The acoustic/vibration sensors may be mounted to the closing wheels or the mounting structure for the closing wheels as indicated by block **280**. The acoustic/vibration sensors may be mounted any other places in order to sense acoustics or vibrations of any type that may be of interest in identifying characteristics. The agricultural planting machine may be configured in a wide variety of other ways with acoustic/vibration sensors and a trained or configured signal processing system, as indicated by block **282** in the flow diagram of FIG. 5.

(63) The agricultural planting machine then begins performing the agricultural operation, as indicated by block **284** in the flow diagram of FIG. 5. During performance of the agricultural operation, the acoustic/vibration sensors generate acoustic/vibration sensor signals **198** which are provided to control system **156**. Generating the acoustic/vibration sensor signals is indicated by block **286** in the flow diagram of FIG. 5. Signal conditioning system **202** may then perform signal conditioning, as indicated by block **288**. The signal conditioning can include filtering **290**, amplifying **292**, sampling **294**, and/or any of a wide variety of other signal conditioning **296**.

(64) Signal processing system **204** then performs signal processing with an advanced signal processor such as a machine learning signal processor (e.g., FFTs, an artificial intelligence system, convolutional neural network, etc.) on the received signals, as indicated by block **298** in the flow diagram of FIG. 5. The received signals can be received at microphone inputs, as indicated by block **300**. The signal processing can include applying FFTs **301**, feature extraction **302** and processing by a convolutional neural network **304**. The advanced processing can take a wide variety of other forms **306** as well.

(65) The acoustic signatures or vibration patterns in the sensor signal can then be identified as, or in response to, extracted features, as indicated by block **308**. The signatures are then classified (or otherwise processed) by the signal processing system **204** in order to identify the processing results, as indicated by block **310**. Signal processing system **204** can use blockage characteristic processor **210** to perform operations that identify blockage characteristics. Such characteristics can include identifying different materials and/or verifying material flow of the different materials expected through the system, as indicated by block **312**. Blockage characteristic processor **210** can verify that the different materials are being delivered through the machine in the expected or desired ratios, as indicated by block **314**. If not, the source of the blockage or partial blockage (e.g., tank, meter, etc.) can be identified. Processor **210** can identify other blockage characteristics as well.

(66) Furrow/soil characteristic processor **212** may perform furrow/soil characterization **316** based upon the sensor signals and the identified acoustic signatures or vibration patterns. Machine characteristic processor **214** can identify machine characteristics **318**, and/or signal processing system **204** can perform a wide variety of other processing **320** as well.

(67) Control signal generator **206** can generate control signals based upon the classified acoustic signatures or vibration patterns, as indicated by block **322** in the flow diagram of FIG. 5. By way of example, signal processing system **204** can geotag the characteristics, as indicated by block **326**. For instance, signal processing system **204** can obtain the geographic position (where the acoustic/vibration signal was generated) from position sensor **194** and correlate that with the characteristics that are identified (the blockage characteristics, the furrow/soil characteristics, the machine characteristics, etc.). The signal processing system **204** can also correlate the identified characteristics to different environmental conditions, material conditions, machine conditions, etc., as indicated by block **328**. The environmental conditions may include such things as soil characteristics, humidity or other weather conditions. The material conditions may include such things as material type, material density, material moisture, etc. The machine characteristics may include such things as machine speed, machine orientation or pose, machine location and heading, air pressures, machine temperature, fan speeds, or any of a wide variety of other machine characteristics.

(68) The control signal can be used to control operator interface system **190** to generate an operator alert, and/or to identify the problem, such as the meter and tank that are blocked, the location of an obstacle or residue, and/or other blockage, furrow/soil, or machine characteristics. Generating an operator alert is indicated by block **324** of FIG. 5.

(69) Control signal generator **206** can generate a control signal to control communication system **184** to communicate the identified characteristics to other systems **224** or other machines **226**, as indicated by block **330** in the flow diagram of FIG. 5.

(70) Control signal generator **206** can generate the control signal to control data store **182** to store the identified characteristic and any other metadata (such as conditions, geographic location, etc.) locally on data store **182** or remotely on another system **224**, as indicated by block **332** in the flow diagram of FIG. 5. Control signal generator **206** can generate the control signal to control the machine (such as a stop signal to stop the machine, a signal to change machine speed or machine heading, a signal to change downforce, etc.) as indicated by block **333**. Signal processing system **204** and control signal generator **206** can perform other processing and generate a wide variety of other control signals as well, as indicated by block **334** in the flow diagram of FIG. 5.

(71) Until the operation is complete, as indicated by block **336** in FIG. 5, processing reverts to block **284** where the agricultural planting machine continues to perform the agricultural operation.

(72) It can thus be seen that the present description describes a system in which acoustic/vibration sensors are used to generate acoustic/vibration signals. Different features, patterns, and/or signatures can be identified in the acoustic/vibration signals and those features, patterns, and/or signatures can be processed by a machine learning or artificial intelligence processing system to identify different blockage characteristics, furrow/soil characteristics, machine characteristics, and/or other characteristics. The identified characteristics can be used to generate control signals.

(73) The present discussion has mentioned processors and servers. In one example, the processors and servers include computer processors with associated memory and timing circuitry, not separately shown. The processors and servers are functional parts of the systems or devices to which they belong and are activated by, and facilitate the functionality of the other components or items in those systems.

(74) Also, a number of user interface displays have been discussed. The displays can take a wide variety of different forms and can have a wide variety of different user actuable input mechanisms disposed thereon. For instance, the user actuable input mechanisms can be text boxes, check boxes, icons, links, drop-down menus, search boxes, etc. The mechanisms can also be actuated in a wide variety of different ways. For instance, the mechanisms can be actuated using a point and click device (such as a track ball or mouse). The mechanisms can be actuated using hardware buttons, switches, a joystick or keyboard, thumb switches or thumb pads, etc. The mechanisms can also be actuated using a virtual keyboard or other virtual actuators. In addition, where the screen on which they are displayed is a touch sensitive screen, they can be actuated using touch gestures. Also, where the device that displays them has speech recognition components, they can be actuated using speech commands.

(75) A number of data stores have also been discussed. It will be noted the data stores can each be broken into multiple data stores. All can be local to the systems accessing them, all can be remote, or some can be local while others are remote. All of these configurations are contemplated herein.

(76) Also, the figures show a number of blocks with functionality ascribed to each block. It will be noted that fewer blocks can be used so the functionality is performed by fewer components. Also, more blocks can be used with the functionality distributed among more components.

(77) It will be noted that the above discussion has described a variety of different systems, components, generators, and/or logic. It will be appreciated that such systems, components and/or logic can be comprised of hardware items (such as processors and associated memory, or other processing components, some of which are described below) that perform the functions associated with those systems, components, generators, and/or logic. In addition, the systems, components,

generators, and/or logic can be comprised of software that is loaded into a memory and is subsequently executed by a processor or server, or other computing component, as described below. The systems, components, generators, and/or logic can also be comprised of different combinations of hardware, software, firmware, etc., some examples of which are described below. These are only some examples of different structures that can be used to form the systems, components, generators, and/or logic described above. Other structures can be used as well.

(78) FIG. 6 is a block diagram of system **100**, shown in FIGS. 1A, 1B and 4A, except that system **100** communicates with elements in a remote server architecture **500**. In an example, remote server architecture **500** can provide computation, software, data access, and storage services that do not require end-user knowledge of the physical location or configuration of the system that delivers the services. In various examples, remote servers can deliver the services over a wide area network, such as the internet, using appropriate protocols. For instance, remote servers can deliver applications over a wide area network and they can be accessed through a web browser or any other computing component. Software or components shown in previous FIGS. as well as the corresponding data, can be stored on servers at a remote location. The computing resources in a remote server environment can be consolidated at a remote data center location or they can be dispersed. Remote server infrastructures can deliver services through shared data centers, even though they appear as a single point of access for the user. Thus, the components and functions described herein can be provided from a remote server at a remote location using a remote server architecture. Alternatively, they can be provided from a conventional server, or they can be installed on client devices directly, or in other ways.

(79) In the example shown in FIG. 6, some items are similar to those shown in previous FIGS. and they are similarly numbered. FIG. 6 specifically shows that system **156** and system(s) **224**, and data store **164** can be located at a remote server location **502**. Therefore, system **100** accesses those systems through remote server location **502**.

(80) FIG. 6 also depicts another example of a remote server architecture. FIG. 6 shows that it is also contemplated that some elements of previous FIGS are disposed at remote server location **502** while others are not. By way of example, data store **164** can be disposed at a location separate from location **502**, and accessed through the remote server at location **502**. Regardless of where the items are located, the items can be accessed directly by system **100**, through a network (either a wide area network or a local area network), they can be hosted at a remote site by a service, or they can be provided as a service, or accessed by a connection service that resides in a remote location. Also, the data can be stored in substantially any location and intermittently accessed by, or forwarded to, interested parties. For instance, physical carriers can be used instead of, or in addition to, electromagnetic wave carriers. In such an example, where cell coverage is poor or nonexistent, another mobile machine (such as a fuel truck) can have an automated information collection system. As the tractor comes close to the fuel truck for fueling, the system automatically collects the information from the tractor using any type of ad-hoc wireless connection. The collected information can then be forwarded to the main network as the fuel truck reaches a location where there is cellular coverage (or other wireless coverage). For instance, the fuel truck may enter a covered location when traveling to fuel other machines or when at a main fuel storage location. All of these architectures are contemplated herein. Further, the information can be stored on the tractor until the tractor enters a covered location. The tractor, itself, can then send the information to the main network.

(81) It will also be noted that the elements of previous FIGS., or portions of them, can be disposed on a wide variety of different devices. Some of those devices include servers, desktop computers, laptop computers, tablet computers, or other mobile devices, such as palm top computers, cell phones, smart phones, multimedia players, personal digital assistants, etc.

(82) FIG. 7 is a simplified block diagram of one illustrative example of a handheld or mobile computing device that can be used as a user's or client's hand held device **16**, in which the present

system (or parts of it) can be deployed. For instance, a mobile device can be deployed in the operator compartment of tractor **106** for use in generating, processing, or displaying the yaw rate or meter control data. FIGS. **8-9** are examples of handheld or mobile devices.

(83) FIG. **7** provides a general block diagram of the components of a client device **16** that can run some components shown in previous FIGS., that interacts with them, or both. In the device **16**, a communications link **13** is provided that allows the handheld device to communicate with other computing devices and under some examples provides a channel for receiving information automatically, such as by scanning. Examples of communications link **13** include allowing communication through one or more communication protocols, such as wireless services used to provide cellular access to a network, as well as protocols that provide local wireless connections to networks.

(84) In other examples, applications can be received on a removable Secure Digital (SD) card that is connected to an interface **15**. Interface **15** and communication links **13** communicate with a processor **17** (which can also embody processors or servers from previous FIGS.) along a bus **19** that is also connected to memory **21** and input/output (I/O) components **23**, as well as clock **25** and location system **27**.

(85) I/O components **23**, in one example, are provided to facilitate input and output operations. I/O components **23** for various examples of the device **16** can include input components such as buttons, touch sensors, optical sensors, microphones, touch screens, proximity sensors, accelerometers, orientation sensors and output components such as a display device, a speaker, and or a printer port. Other I/O components **23** can be used as well.

(86) Clock **25** illustratively comprises a real time clock component that outputs a time and date. It can also, illustratively, provide timing functions for processor **17**.

(87) Location system **27** illustratively includes a component that outputs a current geographical location of device **16**. This can include, for instance, a global positioning system (GPS) receiver, a LORAN system, a dead reckoning system, a cellular triangulation system, or other positioning system. System **27** can also include, for example, mapping software or navigation software that generates desired maps, navigation routes and other geographic functions.

(88) Memory **21** stores operating system **29**, network settings **31**, applications **33**, application configuration settings **35**, data store **37**, communication drivers **39**, and communication configuration settings **41**. Memory **21** can include all types of tangible volatile and non-volatile computer-readable memory devices. Memory **21** can also include computer storage media (described below). Memory **21** stores computer readable instructions that, when executed by processor **17**, cause the processor to perform computer-implemented steps or functions according to the instructions. Processor **17** can be activated by other components to facilitate their functionality as well.

(89) FIG. **8** shows one example in which device **16** is a tablet computer **600**. In FIG. **8**, computer **600** is shown with user interface display screen **602**. Screen **602** can be a touch screen or a pen-enabled interface that receives inputs from a pen or stylus. Computer **600** can also use an on-screen virtual keyboard. Of course, computer **600** might also be attached to a keyboard or other user input device through a suitable attachment mechanism, such as a wireless link or USB port, for instance. Computer **600** can also illustratively receive voice inputs as well.

(90) FIG. **9** shows that the device can be a smart phone **71**. Smart phone **71** has a touch sensitive display **73** that displays icons or tiles or other user input mechanisms **75**. Mechanisms **75** can be used by a user to run applications, make calls, perform data transfer operations, etc. In general, smart phone **71** is built on a mobile operating system and offers more advanced computing capability and connectivity than a feature phone.

(91) Note that other forms of the devices **16** are possible.

(92) FIG. **10** is one example of a computing environment in which elements of previous FIGS., or parts of it, (for example) can be deployed. With reference to FIG. **10**, an example system for

implementing some embodiments includes a computing device in the form of a computer **810** programmed to operate as described above. Components of computer **810** may include, but are not limited to, a processing unit **820** (which can comprise processors or servers from previous FIGS.), a system memory **830**, and a system bus **821** that couples various system components including the system memory to the processing unit **820**. The system bus **821** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. Memory and programs described with respect to previous FIGS. can be deployed in corresponding portions of FIG. **10**.

(93) Computer **810** typically includes a variety of computer readable media. Computer readable media can be any available media that can be accessed by computer **810** and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media. Computer storage media is different from, and does not include, a modulated data signal or carrier wave. It includes hardware storage media including both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computer **810**. Communication media may embody computer readable instructions, data structures, program modules or other data in a transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

(94) The system memory **830** includes computer storage media in the form of volatile and/or nonvolatile memory such as read only memory (ROM) **831** and random access memory (RAM) **832**. A basic input/output system **833** (BIOS), containing the basic routines that help to transfer information between elements within computer **810**, such as during start-up, is typically stored in ROM **831**. RAM **832** typically contains data and/or program modules that are immediately accessible to and/or presently being operated on by processing unit **820**. By way of example, and not limitation, FIG. **10** illustrates operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

(95) The computer **810** may also include other removable/non-removable volatile/nonvolatile computer storage media. By way of example only, FIG. **10** illustrates a hard disk drive **841** that reads from or writes to non-removable, nonvolatile magnetic media, an optical disk drive **855**, and nonvolatile optical disk **856**. The hard disk drive **841** is typically connected to the system bus **821** through a non-removable memory interface such as interface **840**, and optical disk drive **855** are typically connected to the system bus **821** by a removable memory interface, such as interface **850**.

(96) Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (e.g., ASICs), Application-specific Standard Products (e.g., ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

(97) The drives and their associated computer storage media discussed above and illustrated in FIG. **10**, provide storage of computer readable instructions, data structures, program modules and other data for the computer **810**. In FIG. **10**, for example, hard disk drive **841** is illustrated as storing operating system **844**, application programs **845**, other program modules **846**, and program data **847**. Note that these components can either be the same as or different from operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

(98) A user may enter commands and information into the computer **810** through input devices such as a keyboard **862**, a microphone **863**, and a pointing device **861**, such as a mouse, trackball or touch pad. Other input devices (not shown) may include a joystick, game pad, satellite dish, scanner, or the like. These and other input devices are often connected to the processing unit **820** through a user input interface **860** that is coupled to the system bus, but may be connected by other interface and bus structures. A visual display **891** or other type of display device is also connected to the system bus **821** via an interface, such as a video interface **890**. In addition to the monitor, computers may also include other peripheral output devices such as speakers **897** and printer **896**, which may be connected through an output peripheral interface **895**.

(99) The computer **810** is operated in a networked environment using logical connections (such as a controller area network—CAN, local area network—LAN, or wide area network WAN) to one or more remote computers, such as a remote computer **880**.

(100) When used in a LAN networking environment, the computer **810** is connected to the LAN **871** through a network interface or adapter **870**. When used in a WAN networking environment, the computer **810** typically includes a modem **872** or other means for establishing communications over the WAN **873**, such as the Internet. In a networked environment, program modules may be stored in a remote memory storage device. FIG. **10** illustrates, for example, that remote application programs **885** can reside on remote computer **880**.

(101) It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein.

(102) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

Claims

1. An agricultural system, comprising: an agricultural machine configured to deliver seed and material to a field; an acoustic/vibration sensor configured to sense sound or vibration generated by a component of the agricultural machine and to generate a sensor signal responsive to the sensed sound or vibration, wherein the sound or vibration generated by the component is responsive to operation of the agricultural machine, and wherein the component is different than the acoustic/vibration sensor; a signal processing system configured to identify a machine operation characteristic based on the sensor signal; and a control signal generator configured to generate a control signal to control the agricultural machine based on the identified characteristic.
2. The agricultural system of claim 1 wherein the signal processing system comprises: an artificial intelligence processing system.
3. The agricultural system of claim 1 wherein the signal processing system comprises: a convolutional neural network.
4. The agricultural system of claim 1 wherein the component comprises a seeding tool configured to apply the seed to the field, wherein the acoustic/vibration sensor is mounted relative to the seeding tool to sense sound or vibration generated by the seeding tool and wherein the signal processing system comprises: a machine characteristic processor configured to identify, as the machine operation characteristic, a characteristic of the seeding tool based on the sensor signal.
5. The agricultural system of claim 4 wherein the machine characteristic processor is configured to identify, as the characteristic of the seeding tool, a characteristic of the seeding tool indicative of whether a portion of the seeding tool is operating based on the sensor signal.
6. The agricultural system of claim 4 wherein the machine characteristic processor is configured to identify, as the characteristic of the seeding tool, a characteristic of the seeding tool indicative of a

wear level of a portion of the seeding tool based on the sensor signal.

7. The agricultural system of claim 4 wherein the machine characteristic processor is configured to identify, as the characteristic of the seeding tool, a characteristic of the seeding tool indicative of whether a portion of the seeding tool is operating in a predefined way based on the sensor signal.

8. The agricultural system of claim 1 wherein the component comprises a material delivery system configured to deliver the material to the field, wherein the acoustic/vibration sensor is mounted relative to the material delivery system to sense sound or vibration generated by the material delivery system, and wherein the signal processing system comprises: a machine characteristic processor configured to identify, as the machine operation characteristic, a characteristic of the material delivery system based on the sensor signal.

9. The agricultural system of claim 1 wherein the component comprises a soil engaging element configured to engage soil in the field and wherein the acoustic/vibration sensor is mounted relative to the soil engaging element to sense sound or vibration generated by the soil engaging element.

10. The agricultural system of claim 9 wherein the signal processing system comprises: a soil characteristic processor configured to identify, as the machine operation characteristic, a characteristic of soil engaged by the soil engaging element based on the sensor signal.

11. The agricultural system of claim 10 wherein the soil characteristic processor is configured to identify, as the characteristic of the soil engaged by the soil engaging element, a characteristic indicative of a level of residue engaged by the soil engaging element based on the sensor signal.

12. The agricultural system of claim 10 wherein the soil characteristic processor is configured to identify, as the characteristic of the soil engaged by the soil engaging element, a characteristic indicative of the soil engaging element striking a rock based on the sensor signal.

13. The agricultural system of claim 10 wherein the soil characteristic processor is configured to identify, as the characteristic of the soil engaged by the soil engaging element, a characteristic indicative of a type of soil engaged by the soil engaging element based on the sensor signal.

14. The agricultural system of claim 9 wherein the soil engaging element comprises a furrow opener configured to open a furrow in the field into which the material is delivered, wherein the acoustic/vibration sensor is mounted relative to the furrow opener to sense sound or vibration generated by the furrow opener, and wherein the signal processing system comprises: a furrow characteristic processor configured to identify, as the machine operation characteristic, a characteristic of the furrow based on the sensor signal.

15. An agricultural system, comprising: an agricultural machine including an application tool configured to perform an agricultural operation applying material to a field; an acoustic/vibration sensor configured to sense noise or vibrations generated by operation of the application tool and to generate a sensor signal responsive to the sensed noise or vibrations; a signal processing system configured to identify a characteristic based on the sensor signal; and an output signal generator configured to generate an output signal based on the identified characteristic.

16. The agricultural system of claim 15 wherein the application tool comprises a seeding tool wherein the acoustic/vibration sensor is mounted relative to the seeding tool to sense sound or vibration generated by operation of the seeding tool and wherein the signal processing system comprises: a machine characteristic processor configured to identify, as the characteristic, a characteristic of the seeding tool based on the sensor signal.

17. The agricultural system of claim 15 wherein the application tool comprises a soil engaging element configured to engage soil in the field and wherein the acoustic/vibration sensor is mounted relative to the soil engaging element to sense sound or vibration generated by operation of the soil engaging element and wherein the signal processing system comprises: a soil characteristic processor configured to identify, as the characteristic, a characteristic of soil engaged by the soil engaging element based on the sensor signal.

18. A method, comprising: applying material to a field with an agricultural machine; sensing, with a sensor, a sound or vibration generated by a component of the agricultural machine, the

component different than the sensor; generating a sensor signal responsive to the sensed sound or the sensed vibration; processing the sensor signal with a machine learning processing system to identify a characteristic based on the sensor signal; and generating an output signal based on the identified characteristic.

19. The method of claim 18 wherein processing the sensor signal comprises: processing the sensor signal to identify, as the characteristic, at least one of a machine characteristic of the agricultural machine, a furrow characteristic of a furrow opened by the agricultural machine, or a soil characteristic of soil in which the furrow is opened.
